Quantum Computing
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Practice Problems for Midterm 1 (lectures of August 25 – October 8, 2026)

These are practice problems in the style of the likely question types, each with a full solution. They are a guess at the exam, not questions from the professor. Every method used is from the scribes; the lecture date is given to find the matching page. Sections 10 and 11 fill two gaps in the scribes: the proof of the Chernoff bound, and the upper-tail version of the max-load exercise.

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1 Run Deutsch–Jozsa, Bernstein–Vazirani or Simon on a Concrete f

The method is always the same: write the state after every gate. Two facts from the scribes (Sep 24, Sep 29) do all the work:

$$U_f|x\rangle|-\rangle = (-1)^{f(x)}|x\rangle|-\rangle, \qquad H^{\otimes n}|x\rangle = \frac{1}{\sqrt{2^n}} \sum_{y\in\{0,1\}^n} (-1)^{x\cdot y}|y\rangle.$$

Problem 1.1 (Deutsch–Jozsa with $n = 2$)**.** Let $f(x_1x_2) = x_1 \oplus x_2$. Run the Deutsch–Jozsa algorithm on f. What is measured, and what does the algorithm conclude?

Solution.

1. Start in $|00\rangle|1\rangle$. Hadamards on all three qubits give $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)|-\rangle$.
2. The query multiplies each $|x\rangle$ by $(-1)^{f(x)}$. Here $f(00) = 0$, $f(01) = 1$, $f(10) = 1$, $f(11) = 0$, so the state is $\frac{1}{2}(|00\rangle - |01\rangle - |10\rangle + |11\rangle)|-\rangle$.
3. Hadamards on the first two qubits. The amplitude of $|y\rangle$ is $\frac{1}{4}\sum_x (-1)^{f(x)+x\cdot y}$.
4. For $y = 00$ this is $\frac{1}{4}(1 - 1 - 1 + 1) = 0$. For $y = 11$ we have $x \cdot y = x_1 \oplus x_2 = f(x)$, so every term is $+1$ and the amplitude is 1.
5. The measurement gives 11 with probability 1. The outcome is not 00, so f is balanced.

One quantum query was used. A deterministic classical algorithm needs $2^{n-1} + 1 = 3$ queries.

Check with a constant function. For $f \equiv 1$ the state after the query is $-\frac{1}{2}\sum_x |x\rangle = -|+\rangle|+\rangle$, and the Hadamards turn it into $-|00\rangle$. The measurement gives 00, so f is constant. The global sign -1 does not affect the measurement. □

Problem 1.2 (Bernstein–Vazirani with $n = 3$)**.** Let $f(x) = s \cdot x \bmod 2$ with the hidden string $s = 101$, so $f(x) = x_1 \oplus x_3$.

(a) How does a classical algorithm find s?

(b) Run the quantum algorithm.

Solution. (a) Query the unit vectors: $f(100) = 1$, $f(010) = 0$, $f(001) = 1$. The answers are the bits of s, so $n = 3$ queries. Fewer cannot work, since each answer is one bit and s has n bits.

(b) After the first Hadamards and the query, the first register is

$$\frac{1}{\sqrt{8}}\sum_x (-1)^{s\cdot x}|x\rangle = \frac{1}{\sqrt{8}}(|000\rangle - |001\rangle + |010\rangle - |011\rangle - |100\rangle + |101\rangle - |110\rangle + |111\rangle).$$

After the second Hadamards the amplitude of $|y\rangle$ is

$$\frac{1}{8}\sum_x (-1)^{s\cdot x + y\cdot x} = \frac{1}{8}\sum_x (-1)^{(s\oplus y)\cdot x} = \begin{cases} 1 & \text{if } y = s, \\ 0 & \text{if } y \neq s, \end{cases}$$

by the sum-of-signs claim (Sep 29): the sum is 2^n if $s \oplus y = 000$ and 0 otherwise. So the measurement gives $y = 101 = s$ with probability 1, after one query. □

Problem 1.3 (Simon with $n = 3$)**.** The function $f\colon \{0,1\}^3 \to \{0,1\}^3$ is given by the table below. Run Simon's algorithm and find s.

x	000	001	010	011	100	101	110	111
$f(x)$	101	010	000	110	000	110	101	010

Solution.

1. After the Hadamards and the query, the state is $\frac{1}{\sqrt{8}}\sum_x |x\rangle|f(x)\rangle$.
2. Measure the second register. Suppose the outcome is 010. The inputs with $f(x) = 010$ are 001 and 111, so the first register is $\frac{1}{\sqrt{2}}(|001\rangle + |111\rangle)$.
3. Apply $H^{\otimes 3}$. The amplitude of $|y\rangle$ is
$$\frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{8}}\Big((-1)^{001\cdot y} + (-1)^{111\cdot y}\Big) = \frac{1}{4}(-1)^{001\cdot y}\Big(1 + (-1)^{110\cdot y}\Big).$$
4. This is $\pm\frac{1}{2}$ if $110 \cdot y = 0$ and 0 if $110 \cdot y = 1$. So y is one of $000, 001, 110, 111$, each with probability $1/4$. The state is $\frac{1}{2}(|000\rangle - |001\rangle + |110\rangle - |111\rangle)$.
5. Every outcome satisfies $y_1 \oplus y_2 = 0$. The same holds for any other outcome in step 2, because every pair $\{x, x \oplus s\}$ has the same difference.
6. Run the circuit $n - 1 = 2$ times. Suppose the outcomes are $y = 111$ and $y = 001$. They give the equations $s_1 \oplus s_2 \oplus s_3 = 0$ and $s_3 = 0$.
7. Solve: $s_3 = 0$ and $s_1 = s_2$. The solutions are 000 and 110, so the nonzero candidate is $s' = 110$.
8. Check with two classical queries: $f(000) = 101 = f(110)$. So f is 2-to-1 and $s = 110$.

What can go wrong. If the two outcomes are dependent (for example 000 and 111, or 001 twice), the system has more than two solutions and we run again. Two outcomes are independent with probability $\frac{3}{4} \cdot \frac{1}{2} = \frac{3}{8}$: the first must be nonzero (3 of 4), and the second must be outside $\{000, y_1\}$ (2 of 4). This matches the bound of at least $1/4$ from Oct 1.

If f were 1-to-1. Step 2 would leave a single $|x\rangle$, every y would have probability $1/8$, and the check in step 8 would fail, so the algorithm would answer $s = 000$. □

2 Grover with Given N and a

Everything follows from three formulas (Oct 6, Oct 8). N is the number of inputs, a the number of solutions, and k the number of iterations:

$$\sin\theta = \sqrt{\frac{a}{N}}, \qquad \mathbb{P}(\text{success after } k \text{ iterations}) = \sin^2((2k+1)\theta), \qquad k = \left\lfloor \frac{\pi}{4\theta} \right\rfloor.$$

The recipe: compute θ, compute k, plug in. Each iteration adds 2θ to the angle, and we want the angle $(2k+1)\theta$ as close to $90°$ as possible.

Problem 2.1 ($N = 4$, $a = 1$)**.** Find θ, the best k and the success probability. Then verify by tracking the four amplitudes.

Solution.

1. $\sin\theta = \sqrt{1/4} = \frac{1}{2}$, so $\theta = 30^\circ = \pi/6$.
2. $k = \lfloor \pi/(4\theta) \rfloor = \lfloor 1.5 \rfloor = 1$.
3. The success probability is $\sin^2(3 \cdot 30^\circ) = \sin^2 90^\circ = 1$. One query finds the solution with certainty; classically 3 queries are needed in the worst case.

Amplitude check. Start with $\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right)$. The phase flip Z_f negates the solution: $\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2}\right)$, with the solution last. The mean is $\frac{1}{4}\left(\frac{1}{2} + \frac{1}{2} + \frac{1}{2} - \frac{1}{2}\right) = \frac{1}{4}$. Inversion about the mean sends each amplitude α to $2 \cdot \frac{1}{4} - \alpha$, which gives $(0, 0, 0, 1)$. □

Problem 2.2 ($N = 8$, $a = 1$, and overshooting)**.** Find the best k, and the success probability for $k = 1, 2, 3$.

Solution.

1. $\sin\theta = 1/\sqrt{8} \approx 0.3536$, so $\theta \approx 0.3614$ rad $\approx 20.7^\circ$.
2. $k = \lfloor \pi/(4 \cdot 0.3614) \rfloor = \lfloor 2.17 \rfloor = 2$.
3. Plug in:

k	Angle $(2k+1)\theta$	Success probability
0	20.7°	$1/8 = 0.125$
1	62.1°	$25/32 \approx 0.781$
2	103.5°	$121/128 \approx 0.945$
3	144.9°	$169/512 \approx 0.330$

The best is $k = 2$ with probability about 0.945. A third iteration rotates past 90° and the probability drops to 0.33: more iterations are not better. □

Problem 2.3 ($N = 8$, $a = 2$)**.** Same N, two solutions: find θ, k and the success probability.

Solution. $\sin\theta = \sqrt{2/8} = \frac{1}{2}$, so $\theta = 30^\circ$, $k = 1$ and the success probability is $\sin^2 90^\circ = 1$. Only a/N matters, so this is the same as $N = 4$, $a = 1$. Whenever $a = N/4$, one iteration succeeds with certainty. □

Problem 2.4 (Using the wrong a)**.** Let $N = 64$. We run the number of iterations that is right for $a = 1$, but in fact $a = 4$. What is the success probability? What should k have been?

Solution.

1. For $a = 1$: $\sin\theta = 1/8$, $\theta \approx 0.1253$ and $k = \lfloor 6.27 \rfloor = 6$. With $a = 1$ this would succeed with probability $\sin^2(13 \cdot 0.1253) \approx 0.997$.

2. With $a = 4$: $\sin\theta = \sqrt{4/64} = \frac{1}{4}$ and $\theta \approx 0.2527$. After 6 iterations the angle is $13 \cdot 0.2527 \approx 3.285$ rad $\approx 188°$, and $\sin^2 \approx 0.02$.

3. The right choice is $k = \lfloor \pi/(4 \cdot 0.2527) \rfloor = \lfloor 3.11 \rfloor = 3$, with success probability $\sin^2(7 \cdot 0.2527) \approx 0.961$.

So with the wrong a the algorithm fails 98% of the time. This is why an unknown a needs a random k (Oct 8): choose k uniformly from $\left\{1, \ldots, \lfloor 2\sqrt{N} \rfloor\right\}$ and the success probability is at least $3/8$ for every a with $1 \leqslant a \leqslant N/2$ (Problem 14.3). □

Problem 2.5 ($a = N/2$ and $a > N/2$)**.** What does Grover's algorithm do when half or more of the inputs are solutions?

Solution. For $a = N/2$: $\theta = 45°$, and the angles $45°, 135°, 225°, \ldots$ all have $\sin^2 = \frac{1}{2}$. Iterating never helps. For $a > N/2$: $\theta > 45°$, so $k = \lfloor \pi/(4\theta) \rfloor = 0$, and measuring the uniform state directly succeeds with probability $a/N > \frac{1}{2}$. In both cases just sample a random x and check it with one query. □

3 Amplitude Amplification for a Procedure with Success Probability p

Amplitude amplification is Grover with the Hadamards replaced by any measurement-free algorithm $\mathcal{A}$ (Oct 8). If $\mathcal{A}$ succeeds with probability p, then

$$\sin\theta = \sqrt{p}, \qquad Q = -\mathcal{A}\, Z_0\, \mathcal{A}^{-1} Z_f,$$
$$\mathbb{P}(\text{success after } k \text{ rounds}) = \sin^2((2k+1)\theta), \qquad k = \left\lfloor \frac{\pi}{4\theta} \right\rfloor.$$

The cost is $2k + 1$ applications of $\mathcal{A}$ or its inverse: one to prepare the state and two per round.

Problem 3.1 ($p = 1/100$)**.** A quantum algorithm $\mathcal{A}$ outputs a good answer with probability $1/100$, and we can check an answer with f. How many runs of $\mathcal{A}$ are needed classically, and how many with amplitude amplification?

Solution.

1. Classically: run, check, repeat. The number of runs is geometric with mean $1/p = 100$.

2. $\sin\theta = \sqrt{1/100} = 0.1$, so $\theta \approx 0.1002$.

3. $k = \lfloor \pi/(4 \cdot 0.1002) \rfloor = \lfloor 7.84 \rfloor = 7$.

4. The success probability is $\sin^2(15 \cdot 0.1002) = \sin^2(1.503) \approx 0.995$.

5. Cost: $2 \cdot 7 + 1 = 15$ applications of $\mathcal{A}$ or its inverse, and 7 queries to f.

So 15 applications instead of 100. In general the cost is at most $\frac{\pi}{2\sqrt{p}} + 1 = \mathcal{O}(1/\sqrt{p})$ against $1/p$. □

Problem 3.2 ($p = 1/4$)**.** Answer the same question for $p = 1/4$.

Solution. $\sin\theta = \frac{1}{2}$, $\theta = 30°$, $k = 1$, and the success probability is $\sin^2 90° = 1$. Three applications of $\mathcal{A}$ or its inverse give a good answer with certainty. Classically the expected number of runs is 4, and no fixed number of runs gives certainty. □

Problem 3.3 (Short questions)**.**

(a) Why must $\mathcal{A}$ contain no measurement?

(b) How is Grover's algorithm a special case?

(c) What if p is unknown?

Solution. (a) The operator Q uses the inverse of $\mathcal{A}$. A unitary circuit is inverted by running it backwards with each gate inverted. A measurement is not unitary and cannot be undone.

(b) Take $\mathcal{A} = H^{\otimes n}$. Then $\mathcal{A}|0^n\rangle$ is the uniform state, $p = a/N$, and $Q = -H^{\otimes n} Z_0 H^{\otimes n} Z_f$ is the Grover iteration.

(c) The formula for k needs p. The theorem in the scribe assumes that p is known; the unknown case was not covered. □

4 Entangled or Product; Unitary or Not

The test for two qubits. Write the state as $a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle$. Suppose it is a product $(\alpha_0|0\rangle + \alpha_1|1\rangle)(\beta_0|0\rangle + \beta_1|1\rangle)$. Multiplying out gives $a = \alpha_0\beta_0$, $b = \alpha_0\beta_1$, $c = \alpha_1\beta_0$, $d = \alpha_1\beta_1$, and therefore

$$ad = \alpha_0\alpha_1\beta_0\beta_1 = bc.$$

So if $ad \neq bc$ the state is entangled. This is the argument for the Bell state (Sep 15) written once for all states. To show that a state is a product, write down the two factors.

Problem 4.1 (Five states)**.** Which of these states are entangled? Prove each answer.

(a) $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$

(b) $\frac{1}{2}(|00\rangle - |01\rangle + |10\rangle - |11\rangle)$

(c) $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)$

(d) $\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$

(e) $\sqrt{1/3}\,|00\rangle + \sqrt{2/3}\,|01\rangle$

Solution. Compare ad with bc:

State	ad	bc	Answer
(a)	$1/4$	$1/4$	Product: $\lvert+\rangle\lvert+\rangle$
(b)	$-1/4$	$-1/4$	Product: $\lvert+\rangle\lvert-\rangle$
(c)	$-1/4$	$1/4$	Entangled
(d)	0	$-1/2$	Entangled
(e)	0	0	Product: $\lvert0\rangle\left(\sqrt{1/3}\,\lvert0\rangle + \sqrt{2/3}\,\lvert1\rangle\right)$

Full write-up for (c). Suppose the state equals $(\alpha_0|0\rangle + \alpha_1|1\rangle)(\beta_0|0\rangle + \beta_1|1\rangle)$. Then $\alpha_0\beta_0 = \frac{1}{2}$, $\alpha_0\beta_1 = \frac{1}{2}$, $\alpha_1\beta_0 = \frac{1}{2}$ and $\alpha_1\beta_1 = -\frac{1}{2}$. The product of the first and the last is $\alpha_0\alpha_1\beta_0\beta_1 = -\frac{1}{4}$. The product of the middle two is $\alpha_0\alpha_1\beta_0\beta_1 = +\frac{1}{4}$. This is a contradiction, so the state is entangled. □

Problem 4.2 (Is the matrix unitary?)**.** Decide for each matrix whether it is unitary.

$$U = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}, \qquad V = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad W = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Solution. Compute $U^\dagger U$, where † means: transpose, and conjugate every entry.

$$U^\dagger U = \frac{1}{2}\begin{pmatrix} 1 & -i \\ -i & 1 \end{pmatrix}\begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 1-i^2 & i-i \\ -i+i & -i^2+1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

So U is unitary. Do not forget the conjugate: without it the diagonal would be $1 + i^2 = 0$.

V is not unitary: $V|1\rangle = |0\rangle + |1\rangle$ has norm $\sqrt{2}$, and a unitary matrix preserves norms (Sep 15). W is not unitary: $W|0\rangle = W|1\rangle = |+\rangle$, so two orthogonal states are sent to the same state; also $W(|0\rangle - |1\rangle) = 0$.

Quick test. A matrix is unitary exactly when its columns are orthonormal, with the complex inner product. □

Problem 4.3 (A small circuit)**.** Apply H to the first qubit and then CNOT (the first qubit is the control) to $|10\rangle$. Is the result entangled?

Solution.

1. H on the first qubit: $|10\rangle$ becomes $\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \otimes |0\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |10\rangle)$.
2. CNOT flips the second qubit when the first is 1: $\frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$.
3. Test: $ad = -\frac{1}{2}$ and $bc = 0$, so the state is entangled.

The same circuit sends $|00\rangle$ to $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$, $|01\rangle$ to $\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$ and $|11\rangle$ to $\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$. □

5 Random Walk on a Small Graph

The recipe (Sep 8, Sep 10), for a graph with m edges:

$$P_{uv} = \begin{cases} \dfrac{1}{\deg(u)} & \text{if } u \sim v, \\ 0 & \text{otherwise,} \end{cases} \qquad p_{t+1} = p_t P, \qquad \pi_v = \frac{\deg(v)}{2m}.$$

The walk converges to π from every start exactly when the graph is connected and not bipartite. A graph is bipartite exactly when it has no odd cycle.

Problem 5.1 (A triangle with a tail)**.** Vertices $1, 2, 3$ form a triangle and vertex 4 is joined to vertex 3.

(a) Write P.

(b) Find π and verify it.

(c) Does the walk converge?

(d) Starting at vertex 4, compute p_1, p_2, p_3.

Solution. (a) The degrees are $2, 2, 3, 1$, so

$$P = \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ \frac{1}{3} & \frac{1}{3} & 0 & \frac{1}{3} \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

Every row sums to 1.

(b) There are $m = 4$ edges, so $\pi = \left(\frac{2}{8}, \frac{2}{8}, \frac{3}{8}, \frac{1}{8}\right) = \left(\frac{1}{4}, \frac{1}{4}, \frac{3}{8}, \frac{1}{8}\right)$. Verify $\pi P = \pi$ entry by entry:

- Vertex 1: $\pi_2 \cdot \frac{1}{2} + \pi_3 \cdot \frac{1}{3} = \frac{1}{8} + \frac{1}{8} = \frac{1}{4}$.
- Vertex 2: $\pi_1 \cdot \frac{1}{2} + \pi_3 \cdot \frac{1}{3} = \frac{1}{8} + \frac{1}{8} = \frac{1}{4}$.
- Vertex 3: $\pi_1 \cdot \frac{1}{2} + \pi_2 \cdot \frac{1}{2} + \pi_4 \cdot 1 = \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{3}{8}$.
- Vertex 4: $\pi_3 \cdot \frac{1}{3} = \frac{1}{8}$.

(c) Yes. The graph is connected, and it contains a triangle (an odd cycle), so it is not bipartite.

(d) $p_0 = (0, 0, 0, 1)$.

- $p_1 = (0, 0, 1, 0)$: from 4 the walk must go to 3.
- $p_2 = \left(\frac{1}{3}, \frac{1}{3}, 0, \frac{1}{3}\right)$: from 3 each neighbour has probability $\frac{1}{3}$.
- $p_3 = \left(\frac{1}{3} \cdot \frac{1}{2},\ \frac{1}{3} \cdot \frac{1}{2},\ \frac{1}{3} \cdot \frac{1}{2} + \frac{1}{3} \cdot \frac{1}{2} + \frac{1}{3} \cdot 1,\ 0\right) = \left(\frac{1}{6}, \frac{1}{6}, \frac{2}{3}, 0\right)$.

Each p_t sums to 1, as it must (a row stochastic matrix preserves probability vectors). □

Problem 5.2 (A path on three vertices)**.** Consider the path $1-2-3$. Find π. Does the walk from vertex 1 converge?

Solution.

1. The degrees are $1, 2, 1$ and $m = 2$, so $\pi = \left(\frac{1}{4}, \frac{1}{2}, \frac{1}{4}\right)$. Check: $\pi P = \left(\frac{1}{2}\cdot\frac{1}{2},\ \frac{1}{4}+\frac{1}{4},\ \frac{1}{2}\cdot\frac{1}{2}\right) = \pi$.
2. The graph is bipartite, with sides $\{1,3\}$ and $\{2\}$.
3. From $p_0 = (1,0,0)$: $p_1 = (0,1,0)$, $p_2 = \left(\frac{1}{2}, 0, \frac{1}{2}\right)$, $p_3 = (0,1,0)$, and so on.
4. The distribution alternates forever between $(0,1,0)$ and $\left(\frac{1}{2}, 0, \frac{1}{2}\right)$. It does not converge, although π exists and satisfies $\pi P = \pi$.

In terms of eigenvalues: P has the eigenvalues 1, 0 and -1. The eigenvalue -1 is what a bipartite graph produces, and its contribution $(-1)^t$ never dies out. □

Problem 5.3 (Short questions)**.** Find the stationary distribution and say whether the walk converges for

(a) a 4-cycle and a 5-cycle,

(b) any d-regular graph on n vertices,

(c) a disconnected graph.

Solution. (a) In a 4-cycle every degree is 2, so π is uniform, $\left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right)$. An even cycle is bipartite, so the walk from a single vertex does not converge. A 5-cycle is an odd cycle, so the walk does converge, to the uniform distribution.

(b) $2m = nd$, so $\pi_v = d/(nd) = 1/n$: the stationary distribution is uniform.

(c) Each component has its own stationary distribution, so π is not unique and the limit depends on the start. □

6 Tail Bounds in a New Setting

Pick the inequality by what is known about X (Sep 1, Sep 3).

Inequality	What it needs	Bound
Markov	$X \geqslant 0$ and $\mathbb{E}[X]$	$\mathbb{P}(X \geqslant b) \leqslant \mathbb{E}[X]/b$
Chebyshev	$\mathbb{E}[X]$ and $\mathbb{V}[X]$	$\mathbb{P}(\lvert X - \mathbb{E}[X]\rvert \geqslant b) \leqslant \mathbb{V}[X]/b^2$
Chernoff	X is a sum of independent 0/1 variables, $\mu = \mathbb{E}[X]$	exponentially small in μ (three forms below)

$$\mathbb{P}(X \geqslant (1+\delta)\mu) \leqslant e^{-\mu\delta^2/3}, \qquad \mathbb{P}(X < (1-\delta)\mu) \leqslant e^{-\mu\delta^2/2} \quad (0 < \delta < 1),$$
$$\mathbb{P}(X \geqslant R) \leqslant 2^{-R} \quad (R \geqslant 6\mu).$$

For a statement about all n bins, slots or coupons, bound one of them and multiply by n (union bound).

Problem 6.1 (The same event, three bounds)**.** A fair coin is tossed n times and X is the number of heads. Bound $\mathbb{P}(X \geqslant 3n/4)$ with each inequality. Evaluate for $n = 100$.

Solution. $\mathbb{E}[X] = n/2$ and $\mathbb{V}[X] = n \cdot \frac{1}{2} \cdot \frac{1}{2} = n/4$.

1. Markov: $\mathbb{P}(X \geqslant 3n/4) \leqslant \frac{n/2}{3n/4} = \frac{2}{3}$.
2. Chebyshev: $X \geqslant 3n/4$ implies $|X - n/2| \geqslant n/4$, so the probability is at most $\frac{n/4}{(n/4)^2} = \frac{4}{n}$.
3. Chernoff: $3n/4 = \left(1 + \frac{1}{2}\right)\mu$ with $\mu = n/2$ and $\delta = \frac{1}{2}$, so the probability is at most $e^{-(n/2)(1/4)/3} = e^{-n/24}$.

For $n = 100$: 0.667, 0.04 and $e^{-4.17} \approx 0.016$. Markov does not improve with n, Chebyshev improves like $1/n$, and Chernoff improves exponentially. □

Problem 6.2 (n balls into n bins)**.** Throw n balls independently and uniformly into n bins, $n \geqslant 4$. Show that with probability at least $1 - 1/n^2$ no bin receives $3\log_2 n$ balls or more.

Solution.

1. Fix a bin and let X be its number of balls. X is a sum of n independent indicators, each 1 with probability $1/n$, so $\mu = 1$.
2. Take $R = 3\log_2 n$. For $n \geqslant 4$ we have $R \geqslant 6 = 6\mu$, so the third Chernoff form applies.
3. $\mathbb{P}(X \geqslant 3\log_2 n) \leqslant 2^{-3\log_2 n} = 1/n^3$.
4. Union bound over the n bins: $\mathbb{P}(\text{some bin has at least } 3\log_2 n \text{ balls}) \leqslant n \cdot \frac{1}{n^3} = \frac{1}{n^2}$. □

Problem 6.3 (Coupon collector with a different number of trials)**.** There are n coupons and each trial gives a uniformly random one. Bound the probability that some coupon is missing after

(a) $2n \ln n$ trials,

(b) $n \ln n + cn$ trials.

(c) What happens after only n trials?

Solution. A fixed coupon is missed by one trial with probability $1 - 1/n$, so it is missing after m trials with probability $\left(1 - \frac{1}{n}\right)^m \leqslant e^{-m/n}$.

(a) $e^{-2\ln n} = 1/n^2$ per coupon, so at most $n \cdot \frac{1}{n^2} = \frac{1}{n}$ by the union bound.

(b) $e^{-\ln n - c} = e^{-c}/n$ per coupon, so at most e^{-c} in total. For $c = 3$ this is about 0.05.

(c) The expected number of missing coupons is $n\left(1 - \frac{1}{n}\right)^n \approx n/e$, by linearity of expectation. About 37% of the coupons are still missing. □

Problem 6.4 (Boosting a Monte Carlo algorithm by majority vote)**.** An algorithm gives the correct yes/no answer with probability $2/3$. Run it t times independently (t odd) and output the majority answer. How large must t be for an error of at most $1/n$?

Solution.

1. Let X be the number of correct runs: a sum of t independent indicators with $\mu = 2t/3$.
2. The majority is wrong only if $X < t/2$.
3. Write $t/2 = (1-\delta)\mu$: $(1-\delta)\cdot\frac{2t}{3} = \frac{t}{2}$ gives $\delta = \frac{1}{4}$.
4. Lower tail: $\mathbb{P}(X < t/2) \leqslant e^{-\mu\delta^2/2} = e^{-(2t/3)(1/16)/2} = e^{-t/48}$.
5. $e^{-t/48} \leqslant 1/n$ when $t \geqslant 48 \ln n$.

So $\mathcal{O}(\log n)$ repetitions bring the error from $1/3$ down to $1/n$. □

7 Lower Bounds by Adversary Argument and by Yao's Principle

There are two templates. **Adversary** (Aug 25), for deterministic algorithms: answer the queries so that two different inputs stay consistent, and the algorithm must be wrong on one. **Yao** (Sep 1, Oct 1), for randomized algorithms: choose a distribution on inputs and show that every deterministic algorithm does badly on average; then every randomized algorithm does badly on some input.

Problem 7.1 (Membership needs n queries)**.** The scribe shows that finding x needs $n-1$ queries when x is promised to be in the array. Show that deciding whether x is in the array at all needs n queries.

Solution.

1. Fix a deterministic algorithm. The adversary answers every query with "not x".
2. Suppose the algorithm stops after at most $n-1$ queries. Then some slot j was never queried.
3. Two arrays agree with all the answers: one has x in slot j, the other has no x anywhere.
4. The algorithm saw the same answers on both, so it gives the same output on both. It is wrong on one.
5. So a correct algorithm makes n queries.

The promise is what saved one query on Aug 25: with a promise, $n-1$ answers of "no" leave only one possible slot. □

Problem 7.2 (Deutsch–Jozsa needs $N/2+1$ deterministic queries)**.** Prove that a deterministic classical algorithm for the Deutsch–Jozsa problem needs $2^{n-1}+1$ queries.

Solution.

1. Let $N = 2^n$. The adversary answers 0 to every query.
2. Suppose the algorithm stops after at most $N/2$ queries. At least $N/2$ inputs are unqueried.
3. Completion 1: $f = 0$ everywhere, which is constant.
4. Completion 2: $f = 1$ on exactly $N/2$ of the unqueried inputs and 0 elsewhere, which is balanced.
5. Both satisfy the promise and agree with every answer, so the algorithm is wrong on one. □

Problem 7.3 (Randomized search needs about $n/2$ expected queries)**.** Use Yao's principle to show that a Las Vegas algorithm for search makes at least $(n-1)/2$ expected queries on some input.

Solution.

1. Input distribution μ: x is in a uniformly random slot, each with probability $1/n$.
2. Take a deterministic algorithm that is always correct. As long as every answer is "no", its queries follow one fixed order $q_1, q_2, \ldots$
3. It cannot stop while two or more slots are unqueried and x has not been seen (the adversary argument). So if x is in slot q_i with $i \leqslant n-1$, it makes at least i queries, and if x is in the last slot it makes $n-1$.
4. The expected number of queries under μ is at least
$$\frac{1}{n}\left(\sum_{i=1}^{n-1} i + (n-1)\right) = \frac{(n-1)(n+2)}{2n} \geqslant \frac{n-1}{2}.$$
5. By Yao's principle the expected cost of a randomized algorithm on its worst input is at least the expected cost of the best deterministic algorithm under μ. So it is at least $(n-1)/2 = \Omega(n)$. □

Problem 7.4 (The same for Monte Carlo algorithms)**.** Show that a randomized search algorithm with success probability at least $2/3$ on every input needs at least $2n/3 - 1$ queries.

Solution.

1. Use the same distribution μ. Take a deterministic algorithm with T queries, in the fixed order $q_1, \ldots, q_T$ while the answers are "no".
2. It finds x if x is in one of these T slots: probability T/n.
3. Otherwise it has seen T answers of "no" and outputs one fixed slot. That slot is the right one with probability at most $1/n$ overall.
4. So it succeeds with probability at most $(T+1)/n$ under μ.

5. If $T+1 < 2n/3$, every deterministic algorithm with T queries errs with probability more than $1/3$ under μ. By Yao's principle (the form proved on Oct 1) every randomized algorithm with T queries errs with probability more than $1/3$ on some input. □

Problem 7.5 (Simon's problem, in outline)**.** State the hard distribution for Simon's problem and the key estimate. Evaluate it for $n = 10$ and 16 queries.

Solution.

1. Distribution: s is uniform among the $N-1$ nonzero strings, and f is a uniformly random 2-to-1 function with period s.
2. An algorithm learns s only from a collision, two queries x and $x \oplus s$.
3. With k distinct queries the probability of a collision is at most $k(k-1)/N$, since there are $k(k-1)/2$ pairs and each pair rules in one value of s out of at least $N/2$.
4. Without a collision, s is still uniform over at least $N/2$ strings, so a guess is right with probability at most $2/N$.
5. The success probability is at most $(k(k-1)+2)/N$.

For $n = 10$, $N = 1024$ and $k = 16 = \sqrt{N}/2$: at most $(240+2)/1024 \approx 0.24$. So $\sqrt{N}/2$ queries are not enough for success probability $2/3$, and the classical complexity is $\Omega\left(\sqrt{2^n}\right)$. □

8 Leader Election with a Changed Parameter

The algorithm (Sep 8) has two parameters: each node becomes a candidate with probability p, and each candidate contacts k random referees. In class $p = 2\ln n/n$ and $k = \left\lceil \sqrt{n \ln n} \right\rceil$. The whole analysis is four estimates, and a changed parameter just changes the numbers in them. Here R_u is the referee set of node u.

$$\mathbb{P}(\text{no candidate}) = (1-p)^n \leqslant e^{-pn}, \qquad \mathbb{P}(R_u \cap R_v = \emptyset) \leqslant \left(1 - \frac{k}{n}\right)^k \leqslant e^{-k^2/n},$$

$$\mathbb{P}(\text{more than one leader}) \leqslant \binom{n}{2} p^2\, e^{-k^2/n}, \qquad \mathbb{E}[\text{messages}] \leqslant 2k \cdot np.$$

Parameters	No leader	More than one leader	Expected messages
$p = 2\ln n/n$, $k = \left\lceil \sqrt{n \ln n} \right\rceil$ (class)	$\leqslant 1/n^2$	$\leqslant 2\ln^2 n/n$	$\leqslant 4k \ln n$
$p = 3\ln n/n$, same k	$\leqslant 1/n^3$	$\leqslant 9\ln^2 n/(2n)$	$\leqslant 6k \ln n$
same p, $k = \left\lceil 2\sqrt{n \ln n} \right\rceil$	$\leqslant 1/n^2$	$\leqslant 2\ln^2 n/n^4$	$\leqslant 4k \ln n$, with k twice as large
same p, $k = \lceil \sqrt{n} \rceil$	$\leqslant 1/n^2$	bound is $2\ln^2 n/e$, useless	$\leqslant 4k \ln n$
$p = 1/n$, same k	about $1/e$	$\leqslant 1/(2n)$	$\leqslant 2k$

Problem 8.1 ($p = 3\ln n/n$)**.** Redo the analysis with candidate probability $p = 3\ln n/n$ and the same k.

Solution.

1. At least one leader: the candidate with the smallest id always wins at all its referees,

so there is no leader only if there is no candidate. $\mathbb{P}(\text{no candidate}) = (1-p)^n \leqslant e^{-pn} = e^{-3\ln n} = 1/n^3$.

2. Two candidates with a common referee cannot both be leaders. Two referee sets are disjoint with probability at most $e^{-k^2/n} \leqslant e^{-\ln n} = 1/n$, as before.

3. A fixed pair is bad (both are candidates, with no common referee) with probability at most p^2/n. The expected number of bad pairs is at most

$$\binom{n}{2}\frac{p^2}{n} \leqslant \frac{n^2}{2} \cdot \frac{9\ln^2 n}{n^2} \cdot \frac{1}{n} = \frac{9\ln^2 n}{2n}.$$

4. By Markov, $\mathbb{P}(\text{more than one leader}) \leqslant \mathbb{P}(\text{at least one bad pair}) \leqslant \frac{9\ln^2 n}{2n}$.

5. Messages: at most $2k$ per candidate and $3\ln n$ candidates in expectation, so at most $6k\ln n = \mathcal{O}\left(\sqrt{n}\log^{3/2} n\right)$.

More candidates make "no leader" less likely, and "two leaders" and the message count slightly larger. The asymptotics do not change. □

Problem 8.2 (Only $\sqrt{n}$ referees)**.** What breaks if $k = \lceil\sqrt{n}\rceil$, without the logarithm?

Solution.

1. Now $k^2/n \approx 1$, so two referee sets are disjoint with probability about $e^{-1} \approx 0.37$ instead of $1/n$.

2. The bound on the expected number of bad pairs becomes $\frac{n^2}{2} \cdot \frac{4\ln^2 n}{n^2} \cdot e^{-1} = \frac{2\ln^2 n}{e}$, which is larger than 1. Markov says nothing.

3. The algorithm itself fails, not only the proof. Take the two candidates with the smallest ids. With probability about 0.37 they share no referee. Then, unless the second one happened to pick the first as a referee, none of its referees hears a smaller id, and both declare themselves leader.

This is the birthday paradox: two random sets of size k out of n meet reliably only when k^2 is larger than n by a log factor. □

Problem 8.3 (Too few candidates)**.** What happens with $p = 1/n$?

Solution. The expected number of candidates is 1, and $\mathbb{P}(\text{no candidate}) = \left(1-\frac{1}{n}\right)^n$, which tends to $1/e \approx 0.37$. So with constant probability nobody is elected. The factor $\ln n$ in p is what pushes this probability down to $1/n^2$. □

Problem 8.4 (Numbers for $n = 1{,}000{,}000$)**.** Evaluate the class algorithm for $n = 10^6$.

Solution. $\ln n \approx 13.82$.

1. Expected candidates: $np = 2\ln n \approx 27.6$.

2. Referees per candidate: $k = \left\lceil \sqrt{10^6 \cdot 13.82} \right\rceil = 3717$.

3. Expected messages: at most $4k \ln n \approx 205{,}000$. The trivial algorithm, where everyone sends its id to everyone, uses about $n^2 = 10^{12}$.

4. Failure probability: at most $\frac{1}{n^2} + \frac{2\ln^2 n}{n} \approx 10^{-12} + 3.8 \times 10^{-4}$.

5. Rounds: 3. □

9 Deterministic, Randomized and Quantum Query Counts

All entries are from the scribes. $N = 2^n$ is the number of inputs.

Problem	Deterministic	Randomized	Quantum	Lecture
Search, one solution among N	$N-1$	$\Omega(N)$	$\mathcal{O}\left(\sqrt{N}\right)$	Aug 25, Sep 1, Oct 6
Deutsch	2	2	1	Sep 24
Deutsch–Jozsa	$N/2+1$	$\mathcal{O}(1)$ for constant error	1, no error	Sep 24
Bernstein–Vazirani	n	n	1	Sep 29
Simon	$\Theta\left(\sqrt{N}\right)$	$\Theta\left(\sqrt{N}\right)$	$\mathcal{O}(n)$	Sep 29, Oct 1
Collision, random function into $[n]$	—	$\Theta(\sqrt{n})$	later in the course	Sep 3

Problem 9.1 (Is Deutsch–Jozsa an exponential speedup?)**.** The quantum algorithm uses 1 query and a deterministic one needs $N/2+1$. Is this an exponential quantum advantage?

Solution. Only against deterministic (or zero-error) algorithms. A randomized algorithm queries α random inputs and answers "balanced" if two answers differ. It is always right on a constant f. On a balanced f all α answers agree with probability $2 \cdot \left(\frac{1}{2}\right)^{\alpha} = \left(\frac{1}{2}\right)^{\alpha-1}$. With $\alpha = 11$ queries the error is below $1/1000$, for every n. So against bounded-error randomized algorithms the gap is 1 query against a constant. □

Problem 9.2 (Why Simon is different)**.** Explain why Simon's problem gives a real exponential separation.

Solution. The Oct 1 lower bound is for randomized algorithms with bounded error: success probability $2/3$ needs more than $\sqrt{N}/2$ queries. The quantum algorithm needs $\mathcal{O}(n)$ queries. So the gap is $\sqrt{2^n}$ against n even when the classical algorithm may use randomness and may err. A quantum advantage has to be shown against randomized algorithms, because they are the strongest classical model (Aug 25). □

Problem 9.3 (The numbers for $n = 20$)**.** Fill in the query counts for $n = 20$, $N = 1{,}048{,}576$.

Solution. The classical count for Simon lies between the lower bound of Oct 1 and the deterministic algorithm of Sep 29.

Problem	Classical	Quantum
Deutsch–Jozsa, no error	524,289	1
Bernstein–Vazirani	20	1
Simon	more than 512 ($\sqrt{N}/2$); about 1024 suffice up to a constant	21 per attempt: 19 quantum and 2 classical
Search, one solution	1,048,575 deterministic	804 Grover iterations: $\lfloor \frac{\pi}{4} \cdot 1024 \rfloor$

□

Problem 9.4 (One-line reasons)**.** Give a one-line reason for each statement.

(a) Search is only a quadratic speedup.

(b) Bernstein–Vazirani is n against 1.

(c) Simon and collisions have the same square root.

Solution. (a) Grover uses $\mathcal{O}\left(\sqrt{N}\right)$ queries and the scribe notes that $\Omega\left(\sqrt{N}\right)$ quantum queries are necessary, so there is no exponential gain for unstructured search.

(b) Each classical answer is one bit and s has n bits. One quantum query returns all n bits because the Hadamard transform turns the sign pattern $(-1)^{s\cdot x}$ into $|s\rangle$.

(c) Classically s is found only by a collision, and k queries make about $k^2/2$ pairs, so k must reach about $\sqrt{N}$. This is the birthday paradox of Sep 3. □

10 The Chernoff Bound: Proof

The scribe states the Chernoff bounds (Sep 1) without proof. The proof is three ideas: exponentiate, apply Markov, and use independence to turn the expectation of a product into a product of expectations. This is the standard proof; compare with slides 3–7 of the Chernoff deck.

Theorem 1 ▶ Chernoff Bounds

Let $X = X_1 + \cdots + X_n$ with the X_i independent, $X_i = 1$ with probability p_i and 0 otherwise, and let $\mu = \mathbb{E}[X] = p_1 + \cdots + p_n > 0$. Then

1. $\mathbb{P}(X \geqslant (1+\delta)\mu) \leqslant e^{-\mu\delta^2/3}$ for $0 < \delta < 1$,
2. $\mathbb{P}(X < (1-\delta)\mu) \leqslant e^{-\mu\delta^2/2}$ for $0 < \delta < 1$,
3. $\mathbb{P}(X \geqslant R) \leqslant 2^{-R}$ for $R \geqslant 6\mu$.

Proof. Step 1: Markov on e^{tX}. For $t > 0$ the function e^{tx} is increasing, so $X \geqslant a$ is the same

event as $e^{tX} \geqslant e^{ta}$. Since $e^{tX} > 0$, Markov's inequality gives

$$\mathbb{P}(X \geqslant a) = \mathbb{P}\Big(e^{tX} \geqslant e^{ta}\Big) \leqslant \frac{\mathbb{E}\Big[e^{tX}\Big]}{e^{ta}}.$$

Step 2: independence. e^{tX} is the product of the e^{tX_i}, and these are independent, so the expectation factors. Each factor is $\mathbb{E}\Big[e^{tX_i}\Big] = p_i e^t + (1 - p_i) = 1 + p_i(e^t - 1)$. Using $1 + x \leqslant e^x$,

$$\mathbb{E}\Big[e^{tX}\Big] = \prod_{i=1}^{n}\Big(1 + p_i(e^t - 1)\Big) \leqslant \prod_{i=1}^{n} e^{p_i(e^t-1)} = e^{\mu(e^t-1)}.$$

This holds for every real t, positive or negative.

Step 3: choose t. Put $a = (1 + \delta)\mu$ and $t = \ln(1 + \delta)$, which is positive for $\delta > 0$. Then $e^t - 1 = \delta$ and $e^{ta} = (1 + \delta)^{(1+\delta)\mu}$, so for every $\delta > 0$

$$\mathbb{P}(X \geqslant (1 + \delta)\mu) \leqslant \left(\frac{e^\delta}{(1 + \delta)^{1+\delta}}\right)^{\mu}. \tag{1}$$

This is the general form. The three forms of the theorem are simplifications of it.

Step 4: the upper tail for $0 < \delta < 1$. We need $\delta - (1 + \delta)\ln(1 + \delta) \leqslant -\delta^2/3$. Let

$$g(\delta) = \delta - (1 + \delta)\ln(1 + \delta) + \frac{\delta^2}{3}, \qquad g'(\delta) = -\ln(1 + \delta) + \frac{2\delta}{3}, \qquad g''(\delta) = -\frac{1}{1 + \delta} + \frac{2}{3}.$$

We have $g(0) = 0$ and $g'(0) = 0$. Also g'' is negative for $\delta < \frac{1}{2}$ and positive for $\delta > \frac{1}{2}$, so g' first decreases and then increases. Since $g'(1) = \frac{2}{3} - \ln 2 \approx -0.026 < 0$, the function g' starts at 0, goes down, and has not come back up to 0 by $\delta = 1$: $g' \leqslant 0$ on $[0, 1]$. Hence g is decreasing on $[0, 1]$ and $g(\delta) \leqslant g(0) = 0$. Therefore $\mathbb{P}(X \geqslant (1 + \delta)\mu) \leqslant e^{-\mu\delta^2/3}$.

Step 5: the form for $R \geqslant 6\mu$. Write $R = (1 + \delta)\mu$, so $1 + \delta \geqslant 6$. In Equation (1), bound e^δ by $e^{1+\delta}$:

$$\mathbb{P}(X \geqslant R) \leqslant \left(\frac{e}{1 + \delta}\right)^{(1+\delta)\mu} = \left(\frac{e}{1 + \delta}\right)^{R} \leqslant \left(\frac{e}{6}\right)^{R} \leqslant 2^{-R},$$

because $e/6 \approx 0.453 < \frac{1}{2}$.

Step 6: the lower tail for $0 < \delta < 1$. Now take $t < 0$. Then e^{tx} is decreasing, so $X \leqslant a$ is the same event as $e^{tX} \geqslant e^{ta}$, and Steps 1 and 2 give $\mathbb{P}(X \leqslant a) \leqslant e^{\mu(e^t-1)-ta}$. Put $a = (1 - \delta)\mu$ and $t = \ln(1 - \delta) < 0$:

$$\mathbb{P}(X \leqslant (1 - \delta)\mu) \leqslant \left(\frac{e^{-\delta}}{(1 - \delta)^{1-\delta}}\right)^{\mu}.$$

We need $-\delta - (1 - \delta)\ln(1 - \delta) \leqslant -\delta^2/2$. Let $h(\delta) = -\delta - (1 - \delta)\ln(1 - \delta) + \delta^2/2$. Then $h(0) = 0$ and

$$h'(\delta) = -1 + \ln(1 - \delta) + 1 + \delta = \ln(1 - \delta) + \delta \leqslant 0,$$

since $\ln(1 - \delta) \leqslant -\delta$. So $h \leqslant 0$ on $[0, 1)$, and $\mathbb{P}(X < (1 - \delta)\mu) \leqslant \mathbb{P}(X \leqslant (1 - \delta)\mu) \leqslant e^{-\mu\delta^2/2}$. □

What to remember.

- Independence is used exactly once, in Step 2. Without it the proof fails, and only Markov or Chebyshev are available.
- For $\delta \geqslant 1$ use the general form Equation (1) or the 2^{-R} form. The $e^{-\mu\delta^2/3}$ form is only proved for $\delta < 1$.

11 Max Load, Upper Tail: $3n \ln n$ Queries, at Most $9 \ln n$ per Slot

The Sep 1 scribe proves the lower tail (no slot is accessed too rarely, with $24n \ln n$ queries). The version in the professor's own notes is the upper tail with $3n \ln n$ queries, the same number that makes every slot accessed at least once.

Claim 2 (Max Load, Upper Tail)**.** *Random sampling makes $3n \ln n$ queries into an array with $n \geqslant 2$ slots. Then with probability at least $1 - 1/n^2$ no slot is accessed $9 \ln n$ times or more.*

Proof. Fix a slot i and let Y be the number of queries that access it. Y is a sum of $3n \ln n$ independent indicators, each 1 with probability $1/n$, so $\mu = \mathbb{E}[Y] = 3 \ln n$. The threshold is $9 \ln n = 3\mu = (1 + \delta)\mu$ with $\delta = 2$. Since $\delta = 2$ is not below 1, we use the general form Equation (1):

$$\mathbb{P}(Y \geqslant 9 \ln n) \leqslant \left(\frac{e^2}{3^3}\right)^{3 \ln n} = n^{3(2 - \ln 27)} = n^{-3.88\ldots} \leqslant \frac{1}{n^3}.$$

Here $\ln 27 \approx 3.296$, so the exponent is $3 \cdot (2 - 3.296) \approx -3.89$. By the union bound over the n slots,

$$\mathbb{P}(\text{some slot is accessed at least } 9 \ln n \text{ times}) \leqslant n \cdot \frac{1}{n^3} = \frac{1}{n^2}. \qquad \square$$

This is stronger than the $1 - 1/n$ in the professor's notes.

With only the three forms in the scribe. If only the stated forms may be used, take $R = 6\mu = 18 \ln n$ in the third one: $\mathbb{P}(Y \geqslant 18 \ln n) \leqslant 2^{-18 \ln n} = n^{-18 \ln 2} \approx n^{-12.5}$. The union bound gives a failure probability of at most n^{-11}. The constant is 18 instead of 9, but no new formula is needed.

Putting both tails together. By the Sep 1 theorem, $3n \ln n$ queries access every slot at least once with probability at least $1 - 1/n^2$. With Claim 2 and a union bound, with probability at least $1 - 2/n^2$ every slot is accessed at least once and fewer than $9 \ln n$ times.

Numbers for $n = 1000$. $\ln n \approx 6.91$, so the algorithm makes $3n \ln n \approx 20{,}700$ queries. Each slot is accessed about 20.7 times on average. The cap $9 \ln n$ is about 62. The probability that some slot reaches the cap is at most $1000^{-2.89} \approx 2 \times 10^{-9}$.

12 Birthday Paradox and Collisions

Everything here is from Sep 3. There are m people (queries) and n days (values), and X is the number of colliding pairs.

$$\mathbb{E}[X] = \binom{m}{2}\frac{1}{n} = \frac{m(m-1)}{2n}, \qquad \mathbb{P}(\text{no collision}) = \prod_{i=0}^{m-1}\left(1 - \frac{i}{n}\right) \leqslant e^{-m(m-1)/(2n)},$$

$$\mathbb{P}(X = 0) \leqslant \frac{\mathbb{V}[X]}{\mathbb{E}[X]^2} \leqslant \frac{1}{\mathbb{E}[X]}.$$

The first is linearity of expectation, the second is the chain rule with $1 - x \leqslant e^{-x}$, and the third is Chebyshev with $\mathbb{V}[X] \leqslant \mathbb{E}[X]$.

Problem 12.1 (The numbers for $n = 365$)**.**

(a) How many people make the expected number of collisions at least 1?

(b) How many make the probability of a collision at least $\frac{1}{2}$, by the chain-rule bound?

(c) How many does the Chebyshev proof need to guarantee probability $\frac{1}{2}$?

Solution. (a) We need $m(m-1) \geqslant 2n = 730$. Since $27 \cdot 26 = 702$ and $28 \cdot 27 = 756$, the answer is $m = 28$, with $\mathbb{E}[X] = 756/730 \approx 1.04$.

(b) We need $e^{-m(m-1)/730} \leqslant \frac{1}{2}$, that is $m(m-1) \geqslant 730 \ln 2 \approx 505.997$. Since $22 \cdot 21 = 462$ and $23 \cdot 22 = 506$, the answer is $m = 23$. The exact product for $m = 23$ is 0.4927, so a collision has probability 0.507.

(c) Chebyshev gives $\mathbb{P}(X = 0) \leqslant 1/\mathbb{E}[X]$, so we need $\mathbb{E}[X] \geqslant 2$, that is $m(m-1) \geqslant 1460$. Since $38 \cdot 37 = 1406$ and $39 \cdot 38 = 1482$, the answer is $m = 39$. The condition $m \geqslant 2\sqrt{n}+1 \approx 39.2$ of the exercise gives $m = 40$.

The three answers differ (28, 23, 39) but all are about $\sqrt{n}$. Part (a) alone proves nothing about the probability: $\mathbb{E}[X] \geqslant 1$ does not imply that $\mathbb{P}(X \geqslant 1)$ is large. That step needs more than the expectation: either the direct product of part (b), or the variance, which is what the exercise uses with Chebyshev. □

Problem 12.2 (Why Chebyshev and not Chernoff?)**.** X is a sum of indicators X_{ij}. Why does the birthday proof not use the Chernoff bound? Why is Markov not enough?

Solution.

1. Chernoff needs mutually independent indicators. The X_{ij} are not: if persons 1 and 2 share a birthday and persons 2 and 3 share a birthday, then persons 1 and 3 do too. So $\mathbb{P}(X_{12} = X_{23} = X_{13} = 1) = 1/n^2$, not $1/n^3$.

2. They are pairwise independent: for two different pairs, $\mathbb{P}(X_{ij} = X_{kl} = 1) = 1/n^2 = \mathbb{P}(X_{ij} = 1) \cdot \mathbb{P}(X_{kl} = 1)$, whether the pairs share a person or not.

3. Pairwise independence is exactly what the variance needs. The cross terms $\mathbb{E}[X_{ij}X_{kl}] - \mathbb{E}[X_{ij}]\,\mathbb{E}[X_{kl}]$ are all zero, so $\mathbb{V}[X]$ is the sum of the $\mathbb{V}[X_{ij}]$, which is at most $\mathbb{E}[X]$. Chebyshev then applies.

4. Markov gives $\mathbb{P}(X \geqslant 1) \leqslant \mathbb{E}[X]$. That is an upper bound on the probability of a collision, and we want a lower bound.

The rule: Markov needs only $X \geqslant 0$, Chebyshev needs the variance (pairwise independence is enough), and Chernoff needs mutual independence. □

Problem 12.3 (The collision lower bound with numbers)**.** Let f be a random function into $n = 1{,}000{,}000$ values.

(a) Bound the probability that 500 queries find a collision.

(b) In general, bound it for $k = \sqrt{n}/2$ queries.

(c) How many queries are needed for probability $\frac{1}{2}$, and how many are enough?

Solution. By the Sep 3 exercise, k queries find a collision with probability at most $k(k-1)/(2n)$. The reason is that query number s hits one of the $s-1$ earlier answers with

probability at most $(s-1)/n$, and the union bound adds these up.

(a) $\frac{500 \cdot 499}{2{,}000{,}000} \approx 0.125$.

(b) $\frac{k(k-1)}{2n} < \frac{k^2}{2n} = \frac{n/4}{2n} = \frac{1}{8}$, for every n.

(c) Needed: $\frac{k(k-1)}{2n} \geqslant \frac{1}{2}$ means $k(k-1) \geqslant 1{,}000{,}000$. Since $1000 \cdot 999 = 999{,}000$ is too small, $k \geqslant 1001$. Enough: by the birthday exercise, $m \geqslant 2\sqrt{n} + 1 = 2001$ random queries collide with probability at least $\frac{1}{2}$.

So the answer lies between 1001 and 2001 queries: $\Theta(\sqrt{n})$. These are two separate proofs. The upper bound analyses one algorithm (random queries, Chebyshev). The lower bound holds for every algorithm, adaptive or randomized (union bound). □

13 Expectation, Markov and the Weak Law

Three short problems that each use one tool: linearity of expectation (Aug 27), the weak law from Chebyshev (Sep 8), and Markov (Sep 3).

Problem 13.1 (Empty bins)**.** Throw m balls independently and uniformly into n bins.

(a) Find the expected number of empty bins.

(b) Evaluate it for $m = n$, $m = n \ln n$ and $m = 2n \ln n$.

(c) For $m = 2n \ln n$, bound the probability that some bin is empty.

Solution. Let $Y_i = 1$ if bin i is empty and 0 otherwise, and let $Y = Y_1 + \cdots + Y_n$ be the number of empty bins. One ball misses bin i with probability $1 - 1/n$, and the balls are independent, so $\mathbb{E}[Y_i] = \mathbb{P}(Y_i = 1) = \left(1 - \frac{1}{n}\right)^m$.

(a) By linearity of expectation, $\mathbb{E}[Y] = n\left(1 - \frac{1}{n}\right)^m \leqslant ne^{-m/n}$, using $1 - x \leqslant e^{-x}$.

(b) For $m = n$: about n/e, so 37% of the bins are empty (for $n = 100$ the exact value is 36.6). For $m = n \ln n$: at most $ne^{-\ln n} = 1$. For $m = 2n \ln n$: at most $n \cdot n^{-2} = 1/n$.

(c) $Y \geqslant 0$, so Markov gives $\mathbb{P}(Y \geqslant 1) \leqslant \mathbb{E}[Y] \leqslant 1/n$.

The Y_i are not independent (if every other bin is empty, bin i is not), and linearity does not need them to be. Part (c) is the coupon collector bound again: Markov on the number of empty bins gives the same number as the union bound over the bins. □

Problem 13.2 (Estimating the bias of a coin)**.** A coin shows heads with unknown probability p. We toss it n times and estimate p by the fraction of heads. How large must n be so that the estimate is within ε of p with probability at least $1 - \delta$? Evaluate for $\varepsilon = \delta = 0.05$.

Solution. Let $X_i = 1$ if toss i is heads. Then $\mathbb{E}[X_i] = p$ and $\mathbb{V}[X_i] = p(1-p) \leqslant \frac{1}{4}$, because $p(1-p)$ is largest at $p = \frac{1}{2}$. The estimate is the average $\overline{X}_n$ of $X_1, \ldots, X_n$. By the weak law (the Sep 8 exercise, which is Chebyshev applied to the average),

$$\mathbb{P}\Big(\Big|\overline{X}_n - p\Big| \geqslant \varepsilon\Big) \leqslant \frac{\sigma^2}{n\varepsilon^2} = \frac{p(1-p)}{n\varepsilon^2} \leqslant \frac{1}{4n\varepsilon^2}.$$

This is at most δ when $n \geqslant \frac{1}{4\varepsilon^2\delta}$. For $\varepsilon = \delta = 0.05$: $n \geqslant \frac{1}{4 \cdot 0.0025 \cdot 0.05} = 2000$ tosses.

The bound does not need p, which is the point: p is what we are trying to learn. The weak law only needs pairwise independent tosses. The cost of using Chebyshev is the factor $1/\delta$:

for $\delta = 10^{-6}$ it asks for 10^8 tosses. Independent tosses allow Chernoff, where δ enters only as $\ln(1/\delta)$. □

Problem 13.3 (Las Vegas to Monte Carlo)**.** A Las Vegas algorithm $\mathcal{A}$ is always correct and makes T queries in expectation on every input.

(a) Build an algorithm that never makes more than $2T$ queries and fails with probability at most $\frac{1}{2}$.

(b) Get the failure probability down to $1/n$.

(c) Check (a) on random sampling for search.

Solution. (a) Run $\mathcal{A}$ and stop it after $2T$ queries. If it has finished, output its answer, which is correct. If not, output "fail". Let X be the number of queries that $\mathcal{A}$ would make. $X \geqslant 0$ and $\mathbb{E}[X] \leqslant T$, so by Markov $\mathbb{P}(X \geqslant 2T) \leqslant \frac{\mathbb{E}[X]}{2T} \leqslant \frac{1}{2}$. The new algorithm fails only in that case. Stopping after cT queries gives failure at most $1/c$ in the same way.

(b) Markov alone would need $c = n$, that is nT queries. Instead repeat the $2T$-query version $\lceil \log_2 n \rceil$ times with fresh randomness and stop at the first run that finishes. All runs fail with probability at most $\left(\frac{1}{2}\right)^{\log_2 n} = \frac{1}{n}$. The total is at most $2T\lceil \log_2 n \rceil$ queries.

(c) Random sampling with replacement finds x with probability $1/n$ per query, so the number of queries is geometric with mean $T = n$. Markov says that $2n$ queries fail with probability at most $\frac{1}{2}$. The exact value is $\left(1 - \frac{1}{n}\right)^{2n} \leqslant e^{-2} \approx 0.135$.

Markov is weak here, but it used nothing about $\mathcal{A}$ except its expected cost. Part (b) works because a Las Vegas run tells us whether it finished, so failures are detected and never output as answers. □

14 Three More Quantum Questions

Problem 14.1 (Why do we need U_f?)**.** Let $f\colon \{0,1\}^n \to \{0,1\}$.

(a) Why can no quantum circuit map $|x\rangle$ to $|f(x)\rangle$?

(b) Show that $U_f|x\rangle|y\rangle = |x\rangle|y \oplus f(x)\rangle$ is unitary.

(c) How do we read $f(x)$ from it, and where does the sign $(-1)^{f(x)}$ come from?

Solution. (a) Every quantum gate is unitary, so it preserves norms and can be undone (Sep 15). The map $x \mapsto f(x)$ is not one-to-one as soon as two inputs have the same value, and with more than two inputs that always happens. In vectors: if $f(x) = f(x')$ with $x \neq x'$, a linear map T with $T|x\rangle = |f(x)\rangle$ sends $|x\rangle - |x'\rangle$ to 0. That vector has norm $\sqrt{2}$, so T does not preserve norms. This is the Sep 15 argument for XOR, where 00 and 11 both give 0.

(b) U_f sends the basis state $|x\rangle|y\rangle$ to the basis state $|x\rangle|y \oplus f(x)\rangle$. Applying it twice gives $y \oplus f(x) \oplus f(x) = y$, so $U_f^2 = I$. So U_f permutes the 2^{n+1} basis states, its matrix is a permutation matrix, and a permutation matrix is unitary (the Sep 15 argument for CNOT). The trick is that x is kept: from $(x, y \oplus f(x))$ we can recover y, so nothing is lost.

(c) With $y = 0$: $U_f|x\rangle|0\rangle = |x\rangle|f(x)\rangle$, so measuring the last qubit gives $f(x)$. With the last

qubit in $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$:

$$U_f|x\rangle|-\rangle = |x\rangle \cdot \frac{1}{\sqrt{2}}(|f(x)\rangle - |1 \oplus f(x)\rangle).$$

If $f(x) = 0$ this is $|x\rangle|-\rangle$, and if $f(x) = 1$ it is $-|x\rangle|-\rangle$. So $U_f|x\rangle|-\rangle = (-1)^{f(x)}|x\rangle|-\rangle$, the first fact of Section 1.

CNOT is the smallest example: it is U_f for the one-bit function $f(x) = x$. □

Problem 14.2 (Deterministic Simon for $n = 4$)**.** Run the deterministic algorithm of the Sep 29 exercise for $n = 4$.

(a) Which inputs does it query, and how many?

(b) Which two queries collide if $s = 0110$? If $s = 1100$?

(c) How many queries for $n = 20$, against the obvious deterministic algorithm and the quantum one?

Solution. The algorithm splits x into its first $m = \lceil n/2 \rceil$ bits u and its last $n - m$ bits v, and queries every string of the form $(u, 0 \dots 0)$ and every string of the form $(0 \dots 0, v)$.

(a) Here $m = 2$. The strings $u00$ are $0000, 0100, 1000, 1100$. The strings $00v$ are $0000, 0001, 0010, 0011$. The string 0000 is in both, so there are $4 + 4 - 1 = 7$ queries.

(b) $s = 0110$ has $u = 01$ and $v = 10$. The queries 0100 and 0010 have XOR $0110 = s$, so $f(0100) = f(0010)$. $s = 1100$ has $u = 11$ and $v = 00$. The queries 1100 and 0000 have XOR $1100 = s$, so $f(1100) = f(0000)$.

In general every $s = (u, v) \neq 0000$ is the XOR of the two queried strings $(u, 00)$ and $(00, v)$, and these are different because $s \neq 0000$. So a collision always appears, and by the promise any collision x, x' has $x \oplus x' = s$. If no two answers agree, f is one-to-one and $s = 0000$.

(c) The count is $2^m + 2^{n-m} - 1$. For $n = 20$ that is $1024 + 1024 - 1 = 2047$. Querying inputs until two answers agree needs $2^{19} + 1 = 524{,}289$ queries in the worst case. Simon's algorithm makes $n + 1 = 21$.

So the deterministic count is about $2\sqrt{N}$, the same order as the randomized lower bound of Section 7, and the quantum count is about $\log N$. □

Problem 14.3 (Grover when a is unknown)**.** Let $N = 64$, and let the number of solutions a be unknown, with $1 \leqslant a \leqslant 32$.

(a) Why not just run 6 iterations?

(b) State the algorithm of the Oct 8 exercise and its guarantee.

(c) Compute its success probability for $a = 1$ and for $a = 4$.

(d) How do we make the failure probability small?

Solution. (a) Six iterations is right for $a = 1$ (success 0.997), but for $a = 4$ it rotates past the target and succeeds with probability 0.02 (Problem 2.4). The right k depends on a.

(b) Let $K = \lfloor 2\sqrt{N} \rfloor = 16$. Choose k uniformly from $\{1, \dots, 16\}$, make k Grover iterations

on the uniform state and measure. The exercise proves

$$\mathbb{P}(\text{success}) = \frac{1}{K}\sum_{k=1}^{K} \sin^2((2k+1)\theta) = \frac{1}{2} - \frac{\sin(4(K+1)\theta) - \sin 4\theta}{4K \sin 2\theta} \geqslant \frac{1}{2} - \frac{1}{4K \sin 2\theta} \geqslant \frac{3}{8}.$$

(c) For $a = 1$: $\sin\theta = 1/8$ and $\theta \approx 0.1253$. Then $\sin 2\theta \approx 0.2480$, $\sin 4\theta \approx 0.4806$ and $\sin(68\theta) = \sin(8.522) \approx 0.7849$. The probability is

$$\frac{1}{2} - \frac{0.7849 - 0.4806}{64 \cdot 0.2480} \approx \frac{1}{2} - 0.019 = 0.481.$$

For $a = 4$: $\sin\theta = \frac{1}{4}$ and $\theta \approx 0.2527$. Then $\sin 2\theta \approx 0.4841$, $\sin 4\theta \approx 0.8472$ and $\sin(68\theta) = \sin(17.182) \approx -0.9953$. The probability is

$$\frac{1}{2} - \frac{-0.9953 - 0.8472}{64 \cdot 0.4841} \approx \frac{1}{2} + 0.059 = 0.559.$$

Both are above $3/8 = 0.375$. The reason it works: averaged over many values of k, $\sin^2$ of the angle is about $\frac{1}{2}$, whatever θ is. The last inequality of (b) holds because $K \sin 2\theta \geqslant \sqrt{2N} \cdot \sqrt{2a/N} = 2\sqrt{a} \geqslant 2$, so the subtracted term is at most $1/8$. It uses $a \leqslant N/2$.

(d) Check the measured x with one query to f. If it is not a solution, repeat with a new random k. After r rounds the failure probability is at most $\left(\frac{5}{8}\right)^r$, which is below 1% for $r = 10$.

Each round costs at most $16 = 2\sqrt{N}$ queries, so the algorithm is still $\mathcal{O}\left(\sqrt{N}\right)$ without knowing a. □